

RESEARCH ARTICLE

# Long-term effects of prairie restoration on plant community structure and native population dynamics

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The key to restoring degraded grassland habitats is identifying feasible and effective techniques to reduce the negative impacts of exotic species and promote self-sustaining native populations. It is often difficult to extend monitoring of restoration efforts to evaluate long-term success, but doing so is essential to understanding how initial outcomes change over time. To assess how initial treatment effects persist, we revisited degraded patches of Pacific Northwest prairie habitat 6 years after experimental restoration efforts ceased. We evaluated plant community composition to determine the lasting effects of supplemental native seeding and disturbance treatments (burning, mowing, and herbicide to reduce exotic species). We tracked the persistence of seeded species and measured spread of their populations to evaluate suitability of species for restoration and the ability of the habitat to support native plant populations. We found that plots that received supplemental seeding continued to exhibit higher richness of native species than those left unseeded, and that both seeding and disturbance treatments could positively influence native species abundance over the long term. The initially observed effects of disturbance treatments on reducing exotic grass abundance had diminished, highlighting the importance of long-term monitoring and ongoing control of exotic species. Nevertheless, these treatments significantly influenced the population trajectories of 4 out of 8 seeded native species. There was evidence of spatial advance of most seeded species. Results from extended monitoring confirm that dispersal limitation of native species and difficulties maintaining the reduction of exotic grasses continue to be major barriers to success in restoration of invaded grasslands.

**Key words:** invasive grasses, native seeding, plant diversity, population spread, prairies, restoration success, seed limitation

## Implications for Practice

- Long-term monitoring of restoration projects elucidates differences between treatments with persistent versus ephemeral effects, as well as the emergence of delayed responses.
- Supplemental native seeding effectively improves native species richness and abundance over time, confirming the extreme degree of seed limitation in invaded prairies.
- Although mowing, burning, and herbicide can have ephemeral effects on reducing exotic grasses, these management practices can positively influence population recovery trajectories of native species.
- Measurements of population persistence and spread provide a more comprehensive understanding of restoration species' success and suitability within a habitat.

## Introduction

Habitat degradation, fragmentation, and invasion by exotic species are some of the leading causes of the worldwide decline in biodiversity (Pimental et al. 2000; Brudvig 2011; Bullock et al. 2011), and restoring degraded ecosystems is one of the crucial strategies used to slow this loss. Active and adaptive management is necessary to reverse the negative impacts of these

threats and to encourage the development of ecosystems that are more self-sustaining. Barriers to success arise due to ongoing urban and agricultural development (McDonald et al. 2008; Suding 2011), continuous encroachment by non-native species (Fuchs 2001; Didham et al. 2007; Richardson et al. 2007), and the possibility that a changing climate will affect the distributions and interactions of species (Chen et al. 2011; Nathan et al. 2011). Many restoration projects are of relatively short duration (lasting less than 5 years (Ruiz-Jaen & Aide 2005)), with little or no long-term monitoring to evaluate whether treatment effects persist, or if restoration objectives continue to be met. However,

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determining the legacy effects of treatments can provide land managers with valuable information that can confirm the persistence of initial trends, elucidate delayed responses, and allow for the evaluation of reproducing populations (Bash & Ryan 2002; Herrick et al. 2006; Wortley et al. 2013).

Many restoration projects in grassland ecosystems focus on two main goals: reducing the presence of exotic species (e.g. pasture grasses) and maintaining or enhancing the diversity and abundance of desired native species (Corbin et al. 2004; Rook et al. 2011). Restoration strategies, including mechanical, chemical, and biological reduction of non-native species, have resulted in varying levels of success. Coordination of multiple strategies is one key to successful outcomes (Bakker et al. 2003; Corbin et al. 2004; Sinclair et al. 2006; Thorpe & Stanley 2011). It is unlikely that any one strategy will eliminate the need for further management, but there is currently very little information about the temporal scale of removal-treatment effects or about the rate of resurgence of unwanted species. Further, several studies have shown that addition of native seed is critical to enhancing native diversity in degraded and fragmented landscapes with depleted seed banks of native species (MacDougall & Turkington 2005; Kiehl et al. 2010; Brudvig 2011; Stanley et al. 2011a).

Measuring species establishment is an effective way to assess immediate results from supplemental seeding. When temporal and spatial monitoring scales can be extended, additional measurements of population persistence and spread can provide a comprehensive evaluation of the success of species used for restoration purposes. By using direct measurements of target species' population dynamics in both time (change in population size, density, and fecundity) and space (spread), it may be possible to evaluate if, and how, desired species will persist in the long term (Miller & Hobbs 2007). This integration of population ecology and restoration ecology can further develop our understanding of the ecological responses observed after restoration efforts (Montalvo et al. 1997). Because objectives of restoration are often framed on a landscape scale, it is necessary that species used for restoration have the ability to establish and spread beyond the initial point of colonization to improve habitat quality (Kindscher & Tieszen 1998; Neubert & Caswell 2000; Fuchs 2001).

The purpose of this research was to investigate the lasting effects of multiple restoration treatments on plant community richness and structure, as well as to investigate population dynamics and spread of target species in the context of restoration. We resurveyed six degraded prairie sites in the Pacific Northwest (North America) that received experimental restoration treatments and supplemental native seeding between 2005 and 2008. The prairie and oak savanna ecosystem is one of the most endangered ecosystems in both the United States and Canada (Noss et al. 1995; Fuchs 2001). Diversity of native plants in these prairies is decreasing due to continued habitat loss (caused by urban and agricultural development), invasion by non-native species, and the conversion of prairies to woodlands as a result of fire suppression (MacDougall et al. 2004; Hamman et al. 2011). Given the fragmented arrangement of remnant prairie habitat, populations of native plant species

face increased risks related to small patch size and patch isolation, such as dispersal limitation, edge effects (McCune & Vellend 2015), and increased competition (MacDougall & Turkington 2004, 2006). Restoration and preservation efforts in this ecosystem have increased in recent decades as more plant and animal species reach protected status. Recent conservation and restoration efforts have included targeted reduction of exotic plants (Wilson & Clark 2001; Dennehy et al. 2011), reintroducing extirpated species (Lawrence & Kaye 2008; Slater & Altman 2011; Clements 2013), and mimicking historical disturbance regimes (Hamman et al. 2011; Rook et al. 2011).

Objectives of the original study included determining which restoration treatments (with and without seed addition) successfully reduced cover of non-native grasses and increased native species diversity. Initial findings showed that several treatments were effective at decreasing abundance of non-native species but regardless of treatment, supplemental seeding was required for increasing native species richness (Stanley et al. 2010). In order to assess longer-term success, we wanted to know whether desirable trends persisted in order to retrospectively assess the efficacy of management options. We revisited sites in 2014, 7 years after the last seeding and 6 years after the last disturbance treatments, to (1) evaluate the lasting effects of supplemental seeding on native plant richness and abundance; (2) evaluate the lasting effects of disturbance treatments on the cover of exotic grasses; (3) determine the suitability of seeded species based on persistence across sites and disturbance treatments; and (4) examine how the spatial advance of seeded species has contributed to diversity beyond initial points of colonization.

## Methods

### Study Sites

Between 2005 and 2009, a region-wide experiment was undertaken to identify restoration methods that effectively reduce cover of non-native plant species and increase the abundance of native species in the prairie and oak savanna ecosystem of northwestern North America (Stanley et al. 2010, 2011a, 2011b). We selected 10 sites of degraded (invaded) prairie habitat positioned along a 500-km latitudinal gradient ranging from southern British Columbia to Oregon's Willamette Valley (see Table S1 and Fig. S1, Supporting Information, for site details). Sites varied in nutrient and moisture availability and degree of non-native invasion, but commonalities existed in plant species composition (Stanley et al. 2011b). All sites exhibited remnant native populations (with variable species abundance) among high cover of exotic species. Sites at higher latitudes have greater soil moisture, increased organic matter, and less annual precipitation (Richardson et al. 2012).

These 10 sites were later characterized using a site stress index to describe differences in abiotic conditions. To calculate abiotic stress, Richardson et al. (2012) conducted a principal components analysis using site latitudes and measurements of eight abiotic stressors: soil moisture percentage, extractable

soil carbon and nitrogen, concentrations of nitrate and ammonium, surface cover by litter or bare soil, and biomass of soil microbes. Higher values of each environmental stressor translated to higher environmental stress, with the exception of percent cover of bare soil (Reagan 2014). Generally, site stress ratings decreased with increasing latitude; however, the single site in British Columbia, located at the highest latitude, was assigned an intermediate stress rating, in part due to the drier soil moisture regime (due to its geographic position within a rain shadow).

### Restoration Experimental Design (2005–2009)

The experiment used a split plot design with whole plots assigned to one of five disturbance treatments according to a randomized complete block design. Each site (block) contained 20 treatment plots of 5 × 5 m. Drawing upon the findings of previous restoration studies (Tveten 1997; Smith & Knapp 1999; Wilson & Clark 2001), as well as recommendations from land managers, researchers decided upon four experimental treatment combinations (and a control) to be applied over the course of 4 years (2005–2008). The treatments consisted of burning, mowing, grass-specific herbicide (sethoxydim), and broad-spectrum herbicide (glyphosate), plus a control (see Table S2 for detailed descriptions of treatment combinations and timing of applications), and were replicated four times at each site. Within each plot, a split-plot design was used to test the efficacy of post-treatment native seeding. Eight native species characteristic of the prairie/oak savanna ecosystem were chosen for supplemental seeding in 2006 and 2007: *Achillea millefolium*, *Balsamorhiza deltoidea*, *Danthonia californica*, *Eriophyllum lanatum*, *Festuca roemerii*, *Lomatium* spp. (each site receiving one of two congeners: *L. nudicaule* or *L. utriculatum*), *Plectritis congesta*, and *Ranunculus occidentalis*. Seed mixes contained 40% grasses and 60% forbs (by seed count rather than weight), with an approximate total seeding rate of 1,200 seeds/m<sup>2</sup> (see Stanley et al. 2011a [Table 3] for individual species' seeding rates). Vegetation sampling occurred annually in the spring from 2005 to 2009. A 1-m<sup>2</sup> sampling frame was placed in the center of each plot quadrant and species' percent cover was determined by visual estimation. Because experimental management ceased in 2008, some sites have received intermittent treatments that have been applied to the entirety of the experimental block (see Table S3). The periodic management, which involved burning, mowing, or herbicide depending on the site, was targeted at eliminating problematic non-native species. While these periodic treatments may have had an effect on community composition, they were applied uniformly across entire sites, so the differences detected between disturbance treatments and seeding treatments would be attributed to factors of the original restoration experiment.

### Community Composition Sampling (2014)

In 2014, we conducted follow-up monitoring at 6 of the 10 sites (Fig. 1). Sites were chosen based on accessibility and the ability to locate original plot markers. Resurveying occurred between

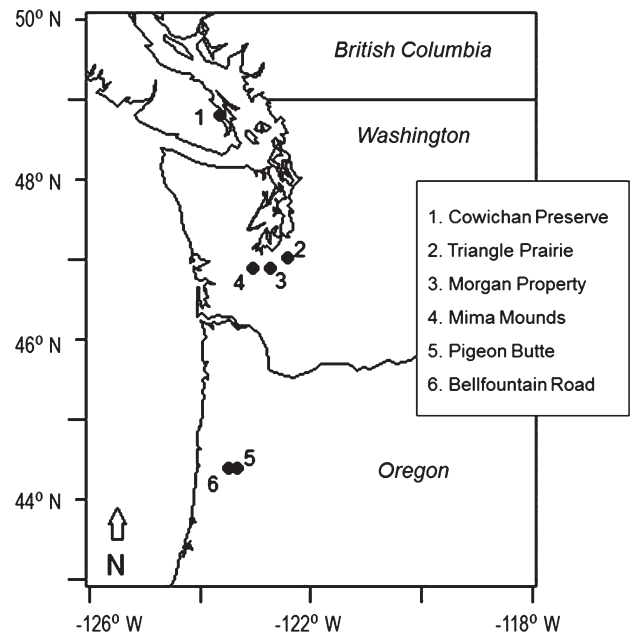


Figure 1. Locations of the six study sites along a 500-km latitudinal gradient in western North America.

April and June to coincide with the sampling window from the original experiment. Community composition was sampled in two 1-m<sup>2</sup> frames within each 5 × 5 m treatment plot: one with no supplemental seed and one seeded in 2006 and 2007. All species present within the sampling frame were identified and assigned a cover value, rounded to the nearest 1%. Total cover in each plot, including substrate and vegetation, was equal to or greater than 100%, depending on the degree of vegetation layering.

### Statistical Analysis

Community data were analyzed using linear mixed-effect models, followed by Tukey's honestly significant difference tests for pairwise comparisons of factor levels. Site, treatment, and seeding with interactions were treated as fixed effects and pre-treatment cover, and count data (from 2005) was included as a covariate to account for initial variance between plots and sites. For analyses comparing seeded and nonseeded subplots, plot was included as a random effect in the model to account for the split-plot design. For each response variable (native species richness, native species abundance, and abundance of exotic grass species), we present results of the full model. Measurements of foliar cover were converted into proportions and transformed using the arcsine-square root. If a site × treatment or site × seeding interaction was significant, analysis was carried out for individual sites. The responses of native seeded species to treatments were analyzed using data from only the seeded plots in order to separate the effect of treatment from the effect of seeding; therefore, fitted models did not require the inclusion of random effects. All analyses were conducted in R 3.1.2 (R Core Development Team, [www.cran-r.org](http://www.cran-r.org)), using packages nlme, lme4, agricolae, and multcomp.

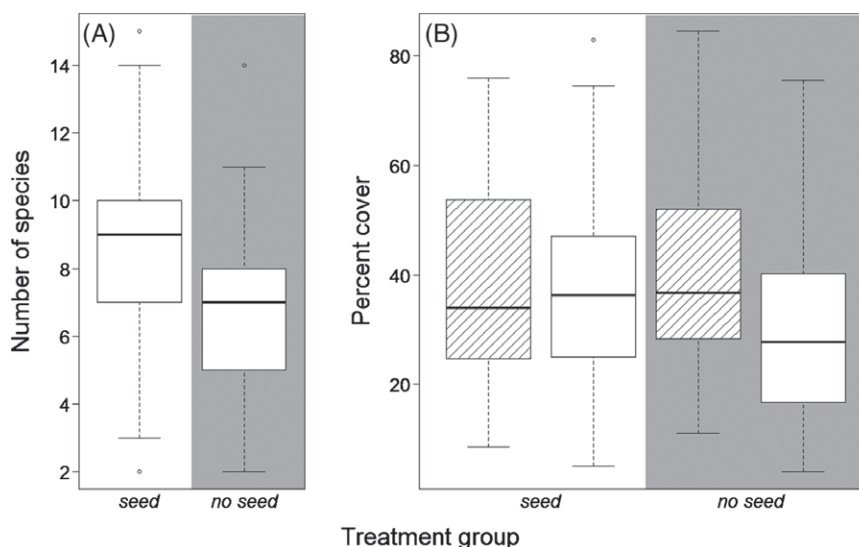


Figure 2. (A) Native species richness and (B) native species abundance (measured by percent cover) in 2014 in plots that were seeded or not in 2006/2007. In panel (B), treatments are broken down further into those with grass-specific herbicide (striped) and those without (white). Median values are represented by the central horizontal line, while upper and lower quartiles are represented by the corresponding ends of the box. Whiskers extend to the most extreme data points excluding outliers, which are represented by open circles.

## Results

### Native Species Richness and Abundance

The number of native species present varied widely across sites, and consistent with original findings, disturbance treatments had no noticeable effect on the number of native species found within plots (i.e. treatment was not statistically significant; Table S4). There was a significant interaction between site and seeding ( $F_{5,89} = 7.11$ ,  $p < 0.001$ ), with all sites except one (Morgan Property) experiencing a positive effect of seeding on native species richness. Across the entire experiment, seeded plots contained an average of 1.92 more native species than unseeded plots (Fig. 2A;  $F_{1,89} = 98.52$ ,  $p < 0.001$ ), down from 3.02 species between the seeding treatments in 2009. Although the addition of native seed effectively increased both the richness and abundance (defined as percent cover) of native species, it was not the sole treatment method that led to an increase in native abundance. In contrast to richness of native species, for which disturbance treatment had no significant influence, the model for predicting abundance included a significant interaction between treatment and seeding (Fig. 2B;  $F_{4,89} = 3.63$ ,  $p = 0.01$ ; Table S4). While all plots receiving supplemental seed had similar average cover of native species regardless of treatment, average cover in unseeded plots differed across treatments. Specifically, when plots were not seeded, the abundance of native species was higher in the two disturbance treatments that included grass-specific herbicide compared with those that did not (Fig. 2B,  $t_{93,8} = -3.05$ ,  $p = 0.003$ ). Site was also a powerful predictor of native abundance ( $F_{5,90} = 40.29$ ,  $p < 0.001$ ).

### Exotic Grass Abundance

Experimental disturbance treatments were designed to remove exotic grasses, without causing major detriment to the native

species present. In 2014, the model predicting cover of exotic grasses (Table S4) included a significant interaction between site and disturbance treatment ( $F_{20,90} = 2.15$ ,  $p = 0.008$ ); therefore, further analysis was conducted at the site level to elucidate treatment effects. Treatment was a significant predictor of non-native grass cover at only one site (Triangle Prairie;  $F_{4,14} = 9.22$ ,  $p < 0.001$ ), where plots that received applications of grass-specific herbicide had lower abundances of exotic grasses than the remaining disturbance treatments. No persistent decline in exotic grasses in response to disturbance treatments occurred at remaining sites. For all sites, seeding was not a significant predictor of exotic grass abundance, so cover values for seeded and unseeded areas in each plot were averaged and the random effect of the split-plot was dropped from the models. Average cover of exotic grasses in 2014 increased from the low levels documented in 2009, but the rate of rebound varied across the range (Fig. 3). Sites in Washington, which experience lower abiotic stress (Richardson et al. 2012), returned to or exceeded 2005 pre-treatment levels, while sites in Oregon and British Columbia, which have intermediate or high abiotic stress, remained below pre-treatment abundance measurements (Fig. 3).

### Persistence and Spread of Seeded Native Species

After five additional years of growth and recruitment, most native seeded species had increased in cover from 2009 to 2014 (examples in Fig. 4). Cover of each native species varied significantly across sites ( $p < 0.01$  for all species), and for four of the eight species, treatment (or a site  $\times$  treatment interaction) was also a significant predictor of changes in abundance since 2005 (Table S5). Specifically, three perennial forb species (*Eriophyllum lanatum*, *Achillea millefolium*, and *Ranunculus*

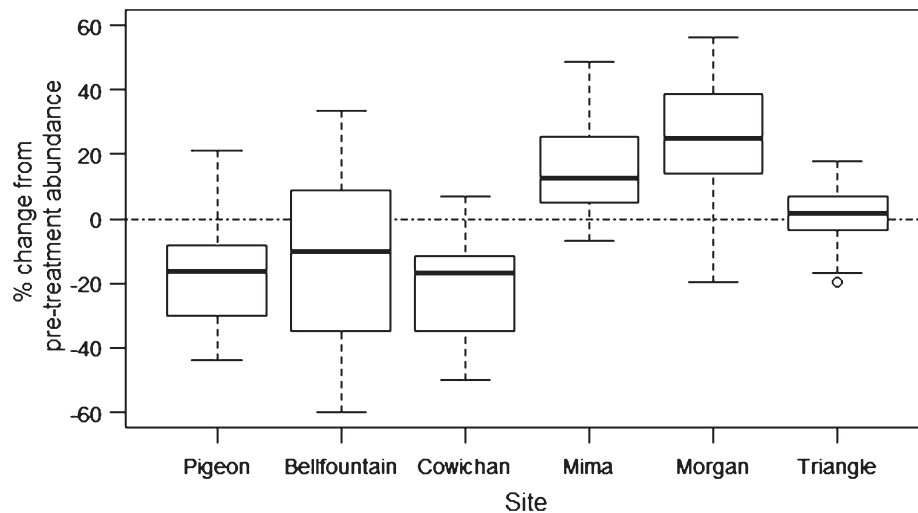


Figure 3. Change in percent cover of exotic grasses from 2005 pre-treatment levels (represented by the dotted line at zero). Sites are arranged in order of decreasing abiotic stress, as defined in Richardson et al. (2012). Median values are represented by the central horizontal line, while upper and lower quartiles are represented by the corresponding ends of the box. Whiskers extend to the most extreme data points excluding outliers, which are represented by open circles.

*occidentalis*) responded more positively to both of the burn treatments compared with other treatments (Fig. 4A–C;  $p < 0.05$ , post hoc Tukey tests). The positive response of *E. lanatum* in the burn plots occurred across all sites, with cover in the burn plots averaging approximately twice that of the control (Fig. 4A). In contrast, the effect of burning on the cover of *A. millefolium* and *R. occidentalis* differed across sites (significant site  $\times$  treatment interaction; Table S5). This result was due to much stronger responses to fire at half of the sites (Pigeon Butte, Mima Mounds, and Triangle Prairie), with little to no noticeable response to treatment elsewhere. Additionally, the native grass species, *Festuca roemerii*, had greater cover in treatments that included the grass-specific herbicide sethoxydim (Fig. 4D;  $p < 0.05$ , post hoc Tukey tests) to which *F. roemerii* is resistant (Sinclair et al. 2006). Neither treatment nor site significantly predicted the presence and cover of the remaining seeded species, *Balsamorhiza deltoidea*, *Lomatium* spp., and *Plectritis congesta*, presumably due to small sample sizes, low persistence, and/or extreme site-specificity.

Although the number of plots with occurrences of the seeded species decreased slightly between 2009 and 2014, there was still a marked increase in frequencies of each of the eight species from pre-treatment conditions (Fig. 5). *Festuca roemerii*, *A. millefolium*, *Danthonia californica*, and *E. lanatum* collectively increased in frequency across all sites since 2009 (Fig. 5), while the remaining four species experienced net declines. While most of these occurrences are of individuals or patches that were present in 2009, there is evidence that all eight species colonized new plots since the last round of monitoring. We found that for six of the seeded species, at least 25% of the currently occupied plots are a result of population spread (as determined by new occupancy in 2014). An annual forb species, *P. congesta*, and a perennial grass species, *D. californica*, had the highest proportions of new occurrences, with 78 and 70% (respectively)

of their current populations, providing evidence of movement beyond initial points of colonization. *Ranunculus occidentalis* and *Lomatium* spp. exhibited the least amount of movement into previously unoccupied sites, but spread still accounted for 17 and 18% (respectively) of their current populations.

## Discussion

Ongoing efforts to restore invaded grasslands face numerous barriers to success due to dwindling native populations (causing seed source limitations) and the high degree of exposure to exotic seed sources (Fuchs 2001; Foster & Tilman 2003). Innovative techniques to decrease the abundance of exotic plant species and increase the richness and abundance of native populations are essential for overcoming these barriers. It is a challenge to understand the trajectories of communities undergoing restoration given the financial and temporal constraints on monitoring. Therefore, the intention of this follow-up study was to determine how the effects of disturbance treatments and native seeding endure over the long term and to evaluate whether short-term results were good predictors of long-term patterns. In summary, these results show that the positive effects of supplementary seeding persist, but that differences observed between exotic removal treatments have diminished over the years. Our results also demonstrate that long-term monitoring of target plant species can elucidate population dynamics that develop slowly over time, and we provide an example of how population spread can be used as a metric to understand the spatial influence of seeding efforts.

## Native Species Richness and Abundance

We found that the early positive effects of supplemental seeding persisted after eight growing seasons (in both disturbance

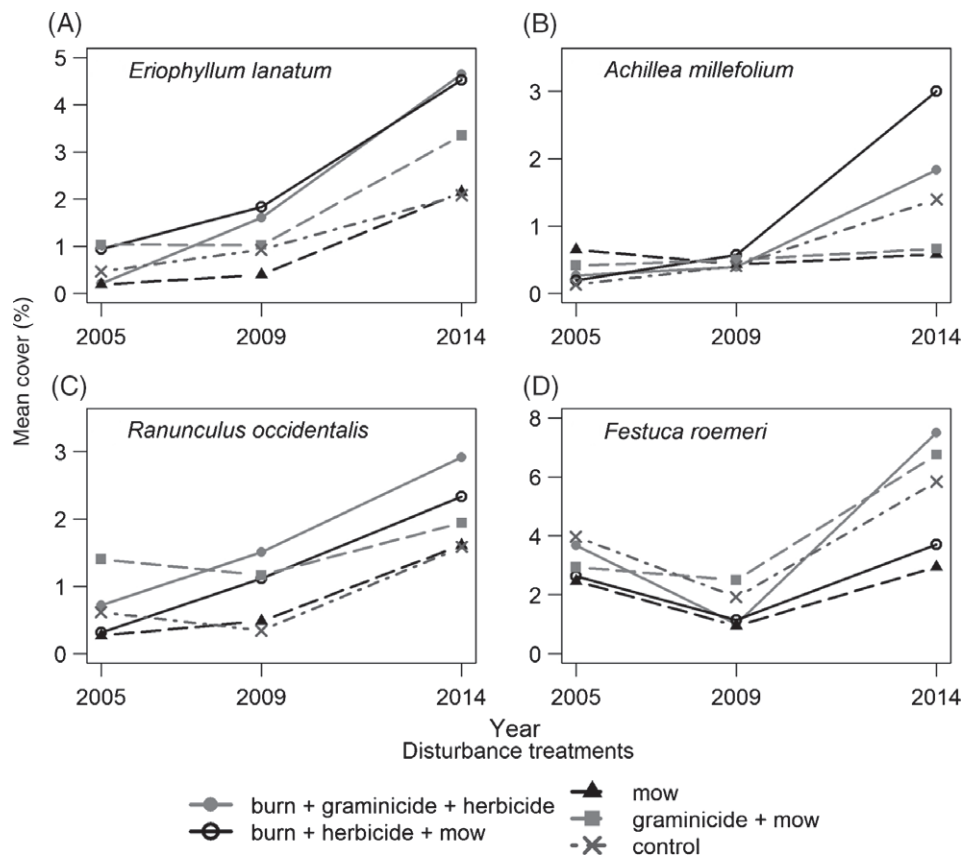


Figure 4. Percent cover of four seeded native species in 2005 (before disturbance treatments were applied), in 2009 (1 year after last disturbance treatments), and in 2014 (6 years after disturbance treatments and seven years after the final seeding). Note different y-axis scales in each panel. "Graminicide" refers to the grass-specific herbicide sethoxydim, while "herbicide" refers to the broad-spectrum herbicide glyphosate.

and control plots), albeit to a slightly lesser degree than was observed immediately after seeding occurred. Positive, lasting effects of seed addition have been seen in a few other grassland habitats (Zobel et al. 2000; Foster & Tilman 2003), but seeding has not been universally successful across all grassland types. Several studies in both arid and subhumid grasslands (Wilsey & Polley 2003; James et al. 2011; Wilson 2015), as well as grass-seeding projects in sagebrush-steppe (Davies et al. 2012) and chaparral (Busby & Southworth 2014), have found that restoration seeding into existing communities often has only ephemeral or marginal effects. Therefore, it is promising to see that seed addition at sites in the prairie/oak savanna ecosystem effectively led to continued improvement in richness of native species and put restoration of the site on a positive trajectory of change. In many cases where persistence of seeded populations has been reported, monitoring was carried out for fewer than five growing seasons (Rinella et al. 2012), but lengthening monitoring periods can highlight differences between immediate and persistent effects of treatments (Peppin et al. 2010; Endress et al. 2012).

This study confirms, once again, that seed limitation is a major barrier to successful restoration of grassland plant communities (Foster & Tilman 2003; Stanley et al. 2010; Richardson et al. 2012). Previous studies have suggested that the relative

importance of seed limitation is likely overestimated when evaluated over short time periods due to the exclusion of factors that affect the fate of populations at later life-stages (Zobel & Kalamees 2005; Clark et al. 2007; Foster et al. 2007; Carrington 2014). The successful establishment of seeded species across all disturbance treatments provided strong evidence for high seed limitation (Reagan 2014), and the persistence of most of these species after 8 years reinforces that seed availability is likely the most limiting factor, above other factors such as negative interactions (e.g. competition for resources, herbivory). If these other factors limited plant reproduction, establishment, and growth, we would expect to see lower establishment from seed or a strong decline in the effect of seeding over time. We found that the overall positive effect of seeding was still noticeable and is unrelated to disturbance treatment; therefore, we conclude that the relative importance of seed limitation was correctly stated at the end of the original monitoring period (Stanley et al. 2011a, 2011b).

A major question posed during the original experiment was whether disturbance treatments alone were sufficient to generate an increase in the richness and abundance of native species, or whether supplemental seeding was required. As discussed earlier, achieving increased richness did, in fact, rely on seeding. Increased abundance of native species, however, was achieved

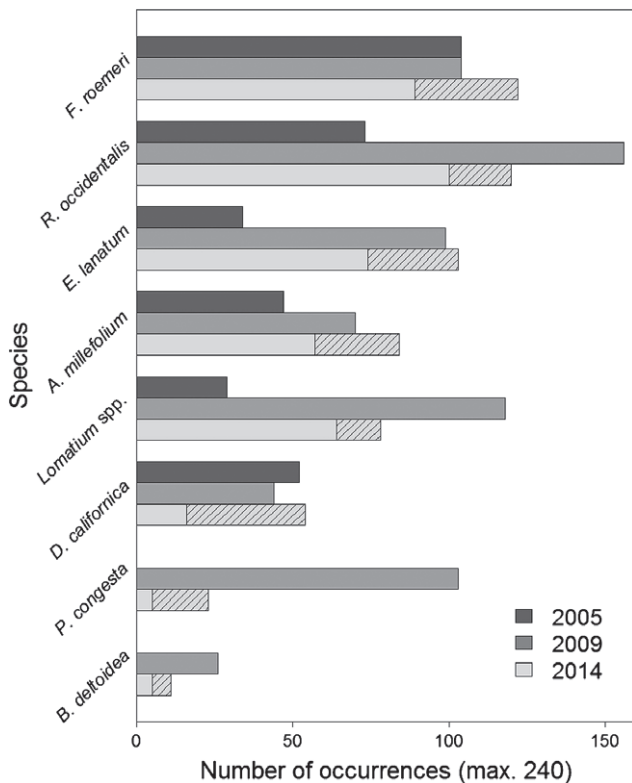


Figure 5. Frequency of seeded native species within monitoring plots before treatment (2005), immediately after treatment (2009), and 5 years after monitoring ceased (2014). Striped portions of the 2014 data indicate occurrences that are a result of population spread (defined as new occupancies since 2009). The maximum number of plots each species could occupy is 240 (6 sites  $\times$  20 plots  $\times$  2 sub-plots). Congeneric *Lomatium* species are grouped together.

either through native seed supplementation or by using a disturbance treatment that included grass-specific herbicide (in the absence of seed supplementation). In the plots treated with grass-specific herbicide, native cover consisted of equal parts grasses (predominantly *Festuca roemerii*) and forbs, indicating that the grass-specific herbicide opened up space for multiple functional groups. Three perennial forb species, *Camassia quamash*, *Fragaria virginiana*, and *Sanicula crassicaulis* (none of which were seeded), responded particularly well to the removal of exotic grasses, likely due to a combination of individual growth and vegetative spread, as well as possible new recruitment, providing some explanation for the increased abundance of native species in the absence of seed supplementation. While seeding is a consistently successful option for increasing cover of native species, the targeted removal of exotic grasses can create sufficient space and resources for existing native plants to expand spatially and to support recruitment of new individuals. This result was not immediately apparent when monitoring ended in 2009, presumably due to a lag time between the removal of exotic grasses and the resulting response of the extant natives being released from competition. In contrast to the original 2009 conclusions, we have now found that supplemental seeding and grass-specific herbicide treatments were equivalent

determinants of increased abundance of native species over the long term, but that their effects were not additive.

### Exotic Grass Abundance

The effects of disturbance treatments on the reduction of non-native grasses were less pronounced in 2014, 5 years after original monitoring had ceased. Initially, treatments that included grass-specific herbicide were the most effective at reducing the abundance of exotic grasses throughout the ecoregion (Stanley et al. 2010), but in 2014 this effect was only detectable at one site. The persistence and dominance of exotic grass species are major barriers to success in grassland restoration and results from this study highlight the extreme difficulty in combating their influence over the long term. Given the aggressive nature of the exotic grasses in this ecosystem, it is unsurprising that there would be a resurgence in abundance in the years following treatment applications, especially given that treatments were not applied to buffer zones between plots, and recolonization through vegetative reproduction or seeding from the edges was likely. Interestingly, the degree of the undesirable rebound differed greatly between sites, with those in Oregon and British Columbia remaining well below pre-treatment levels and those in Washington equaling or exceeding their original levels. Some studies have suggested that successful reduction of exotic species abundance may be associated with areas experiencing high abiotic stress, due to physiological constraints on growth or the lack of traits required for tolerating stressful conditions (Holway et al. 2002; Williamson & Harrison 2002; Gerhardt & Collinge 2003).

The results of this study align with these previous suggestions given that the sites in Washington experience lower abiotic stress than the sites in Oregon and British Columbia (Richardson et al. 2012). Throughout the entirety of the experiment and extended monitoring (2005–2014), Washington sites consistently received higher rainfall and had fewer extreme drought years than both Oregon and British Columbia, potentially facilitating abundant regrowth of exotic grass species that thrive in low-stress conditions. Furthermore, it is worth noting that the two sites that experienced the greatest resurgence of non-native grasses had the lowest initial native species richness (Stanley et al. 2010). There is substantial evidence that more diverse communities have greater resistance to invasion (or in this case, re-invasion) at local scales due to competitive exclusion by established groupings of native species (Tilman 1997; Levine 2000; Naeem et al. 2000). The long-term trend seen in this study elucidates the important roles that both abiotic and biotic factors play in ensuring resistance to invasion within this ecosystem.

### Persistence and Spread of Seeded Native Species

The long-term success of each of the eight seeded species varied considerably. Several species experienced steady, gradual increases in abundance over time (*Achillea millefolium*, *Eriophyllum lanatum*, *Ranunculus occidentalis*); others established well but experienced drastic population declines (*Lomatium utriculatum*, *Plectritis congesta*), while yet another rebounded

after an initial setback (*Festuca roemerii*). When framing restoration goals in the context of habitat enhancement, understanding the mechanisms that support the persistence and spread of desirable plant populations becomes increasingly important. In this experiment, half of the seeded species have the potential to create valuable habitat for threatened Lepidoptera sp., but only if the plant populations can persist to provide nectar sources and larval host plants for multiple generations. Therefore, it is important to determine the optimal restoration techniques (in this case through disturbance treatments) that encourage establishment, but that also have an influence on long-term population persistence and can set the stage for spatial advance.

While forbs native to the oak savanna ecosystem typically establish well following fire (Maret & Wilson 2000; Dunwiddie et al. 2014), it is often unclear whether fire plays a role in determining the potential for a species' persistence. For example, *P. congesta* (the only annual species seeded in this experiment) had the highest establishment success overall (Stanley et al. 2010), particularly in burn treatments, but there was very low persistence after eight growing seasons, and where persistence did occur, it was unrelated to initial disturbance treatment. *A. millefolium*, *E. lanatum*, and *R. occidentalis* (all perennial species) also had greater initial establishment success in plots with a burn treatment (Stanley et al. 2010), but, in contrast with *P. congesta*, the long-term persistence and abundance of these species was positively associated with burn treatments. These contrasting trends between life forms contribute to the recently studied issue of loss of native annual forbs throughout the prairie system in the absence of fire (Dunwiddie et al. 2014). If fire is indeed what promotes establishment of many forbs, the perennial species likely gain enough benefit from a single burn (after which point they can persist relatively well without disturbance), while the viability of annual forb populations remains more deeply dependent on the repeated influence of fire. Unlike the forbs mentioned earlier, the native grass, *F. roemerii*, initially had a negative response to fire followed by glyphosate, as it experiences temporary setbacks after major disturbance (Dunwiddie 2002). In the long term, the most important determinant of success for this species was the reduction of competition with exotic grasses resulting from the use of a grass-specific herbicide, and ultimately it rebounded to higher levels of cover across sites than were initially observed. This difference suggests that the seeded forbs required the creation of suitable microsites to aid establishment (as fire can clear away litter layers, facilitating seedling survival (Richardson et al. 2012)), while *F. roemerii* likely required release from direct competition to aid growth and recruitment.

In addition to measurements of establishment and persistence, using population spread as a metric of success is useful for evaluating longer-term suitability of restoration species and the ability of habitats to support native plant populations. Supplementing sites with native seed can be a cost-prohibitive restoration technique, especially when seed must be locally sourced, so it is essential to understand how seeding efforts influence native species richness and abundance both within and beyond plot boundaries. There is a larger return on seeding investments if the species can successfully reseed and spread beyond the initial

point of application instead of remaining restricted to areas that were directly manipulated. The results of this study show that establishment rates are good metrics for understanding how dispersal and/or microsite availability may limit recruitment, but they are not necessarily indicative of long-term persistence and spread capabilities. We found evidence of movement and colonization by all seeded species, as individuals were observed in plots that had been vacant prior to 2014. Most species seeded in this experiment are wind-dispersed or autodispersed, with only a few having specialized seed dispersal characteristics (*Danthonia californica*, twisted awns; *R. occidentalis*, hooked achenes) or the ability to spread laterally by rhizomes (*A. millefolium*). It is worth noting that the sole annual species, *P. congesta*, differed from most of the perennial species as it had the highest proportion of newly occupied plots but also had the sharpest decline in total occurrences, indicating that the population experienced a high degree of movement, as opposed to spatial expansion. In contrast, the perennial species, *F. roemerii* and *E. lanatum*, had lesser degrees of spread, but the spread led to a net spatial expansion of the population given that the majority of previous occurrences successfully persisted over time. This, once again, highlights that using a combination of establishment, persistence, and spread metrics will provide the most comprehensive evaluation of a species' suitability for restoration purposes.

Seeding native species was the most effective long-term treatment option for increasing native richness across the entire range of sites, while the initial effects of disturbance treatments on reducing the abundance of exotic species diminished over time. It was promising to see that even at sites where the abundance of exotic grasses had rebounded to pre-treatment levels, the populations of seeded species persisted among the invaders. These findings reinforce the major challenge posed by seed limitation in the invaded remnant patches of prairie and oak savanna habitat, but they also provide hope for success in situations where enhancing native diversity is the primary goal (as is often the case with habitat enhancement for sensitive butterfly species).

Collaboration between scientists and land managers provides valuable guidance for habitat restoration practices, and can help managers understand how initial monetary investments relate to long-term improvement. Monitoring the trajectories of the seeded populations, through measurements of persistence and observations of spatial advance, can inform future selection of restoration species and influence seeding plans. Additionally, understanding which restoration techniques produce ephemeral effects (such as the exotic-removal treatments in this experiment) aids in determining appropriate disturbance intervals for ongoing management. Extended monitoring of this restoration experiment has proven beneficial as it revealed the emergence of delayed responses and allowed for evaluation of long-term suitability and success of restoration species.

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## Supporting Information

The following information may be found in the online version of this article:

**Table S1.** Study site details

**Table S2.** Schedule of experimental restoration treatment applications.

**Table S3.** Management treatments applied between 2009 and 2014.

**Table S4.** Results of linear mixed-effect models showing differences in richness and abundance of native species, as well as abundance of exotic grasses, in response to site, disturbance treatment, seeding, and interactions.

**Table S5.** Results of linear models showing differences in abundance of the four native seeded species for which treatment or a site × treatment interaction were significant predictors.

**Figure S1.** Annual temperature and precipitation trends for the two sites in Oregon (Bellfountain Prairie, Pigeon Butte), three sites in Washington (Mima Mounds, Morgan Property, Triangle Prairie), and single site in British Columbia (Cowichan Preserve).

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